Convexifying Contacts in Factor Graph Smoothing for Legged Robots

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***Abstract*—This paper presents a convex optimization framework for contact sequence refinement in factor graph-based state estimation for bipedal robots. Legged robots rely on contact measurements as a key sensory modality, treating foothold positions as landmarks in a factor graph optimization. However, inaccurate contacts and foot slip violate the zero-velocity assumption of the foot in contact which is critical for accurate state estimation. We propose an outer-loop convex optimization method that refines contact sequences by penalizing velocity residuals above a threshold weighted by contact probability, while preserving temporal smoothness. The contact refinement problem is formulated as a Linear Program (LP) with continuous relaxation of binary contact states. Implemented on top of GTSAM's fixed-lag smoother, the approach demonstrates significant improvements in state estimation accuracy on bipedal robot datasets (Unitree G1). Results show 39.9% average reduction in position error and 22.5% improvement in velocity consistency compared to baseline factor graph smoothing. Against simple threshold-based methods, improvements reach 84.3% in position accuracy and 51.8% in velocity consistency.**

***Index Terms*—convex optimization, factor graph optimization, state estimation, bipedal locomotion, contact detection, legged robotics**

I. Introduction

State estimation for legged robots is a challenging problem due to the complex contact dynamics between the robot and its environment. Bipedal and quadrupedal robots experience intermittent contact phases where accurate knowledge of contact timing and locations is critical for estimating the robot's pose and velocity.

Factor graph optimization has emerged as a powerful framework for sensor fusion in robotics, enabling the integration of heterogeneous measurements such as IMU data, kinematic constraints, and contact information [1], [2]. In legged robotics, contact measurements provide kinematic constraints relating the robot body to fixed landmarks (footholds) in the world frame.

Some robots like the Agility Robotic's Cassie have a mechanical contact sensor that provides reliable contact state information. However, many bipedal robots (e.g., Unitree G1) lack such sensors and rely on indirect methods for contact detection. In practice, contact detection algorithms are based on force thresholds, kinematic analysis, or learned methods and are often noisy or have incorrect contact sequences. Furthermore, even when contact is correctly identified, foot slip violates the static foothold assumption, introducing errors into the factor graph. These issues motivate the need for contact sequence refinement.

This paper presents an outer-loop convex optimization approach that refines contact sequences by minimizing foot velocity violations and encouraging temporal smoothness.

Contributions: Our main contributions are: (1) A convex LP formulation for contact sequence refinement with velocity-based objectives, (2) Integration of the refinement loop with GTSAM's fixed-lag smoother framework, and (3) Experimental validation on real bipedal robot data demonstrating improved accuracy.

II. Problem Description

A. Factor Graph Smoothing for Bipedal Robots

We formulate the state estimation problem as maximum a posteriori (MAP) inference on a factor graph. The robot state $\mathbf{x}_t \in SE(3) \times \mathbb{R}^6$ at time t consists of the robot pose (position and orientation) and velocity:

$$\mathbf{x}_t = [\mathbf{R}_t, \mathbf{p}_t, \mathbf{v}_t] \in SE(3) \times \mathbb{R}^3 \tag{1}$$

where $\mathbf{R}_t \in SO(3)$ is the orientation, $\mathbf{p}_t \in \mathbb{R}^3$ is the position, and $\mathbf{v}_t \in \mathbb{R}^3$ is the velocity in the world frame.

B. Contacts as Landmark Measurements

In legged robot state estimation, foot contacts provide useful information about the robot's motion. The key idea is that when a foot is firmly in contact with the ground, its position in the world frame should remain approximately fixed. Therefore, a contact point can be treated similarly to a landmark in SLAM: the robot observes the same fixed point from different body poses.

Let $\mathbf{x}_t$ denote the robot state at time t, which includes the body position $\mathbf{p}_t$ and orientation $\mathbf{R}_t$. When foot i is in contact, we assume that the corresponding foothold has a fixed world-frame position $\mathbf{l}_i \in \mathbb{R}^3$. The robot can measure the position of this foot relative to its body frame through forward kinematics. This gives the measurement model

$$\mathbf{z}_t^i = \mathbf{h}(\mathbf{x}_t, \mathbf{l}_i) + \mathbf{n}_t = \mathbf{R}_t^T(\mathbf{l}_i - \mathbf{p}_t) + \mathbf{n}_t, \tag{2}$$

where $\mathbf{z}_t^i$ is the measured foot position in the body frame, and $\mathbf{n}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}_{\text{contact}})$ represents measurement noise.

This equation has a simple interpretation. If the robot pose $\mathbf{x}_t$ and the foothold position $\mathbf{l}_i$ are correct, then transforming the world-frame foothold into the robot body frame should match the measured foot position from kinematics. Any mismatch becomes a residual that the estimator tries to minimize.

The estimation problem can then be written as a maximum a posteriori (MAP) problem. In MAP estimation, we seek the most likely robot trajectory and foothold positions given all available measurements:

$$\mathbf{x}^*_{0:T}, \mathbf{l}^*_{1:F} = \arg\max_{\mathbf{x}_{0:T}, \mathbf{l}_{1:F}} p(\mathbf{x}_{0:T}, \mathbf{l}_{1:F} \mid \mathbf{z}_{0:T}, \mathbf{u}_{0:T}), \tag{3}$$

where $\mathbf{x}_{0:T}$ is the full robot trajectory, $\mathbf{l}_{1:F}$ are the unknown foothold positions, $\mathbf{z}_{0:T}$ are the sensor measurements, and $\mathbf{u}_{0:T}$ are the motion inputs or inertial measurements.

Equivalently this MAP problem becomes a nonlinear least-squares problem. The estimator searches for the trajectory and foothold locations that best satisfy all measurement constraints. IMU measurements encourage smooth and dynamically consistent motion between time steps, while contact measurements encourage the contacted foot to remain fixed in the world frame.

Thus, the contact measurement acts as an additional source of information that reduces drift from the IMU. Instead of only relying on the IMU, the estimator also uses the fact that a stance foot should not move relative to the ground. This makes foot contact especially valuable for legged robots, because it provides a temporary "anchor" to the world whenever a foot is in contact.

C. The Problem: Faulty Contact Sequences

The factor graph formulation assumes that contact states $c_t^i \in \{0, 1\}$ (binary: contact or no contact) are known a priori. In practice, these are estimated using:

- Force/torque sensor thresholds
- Ground reaction force estimates from kinematics
- Heuristic rules based on gait phase
- Learned classifiers using IMU and joint data (which we developed for Unitree G1 and will be using as the baseline contact detector)

These methods produce noisy contact sequences with two main failure modes:

1) **False positives**: Incorrectly detecting contact during swing phase
2) **False negatives**: Missing contact during stance phase
3) **Foot slip**: Even with correct contact detection, the foot may slide on the ground, violating the zero-velocity assumption

When contact state $c_t^i = 1$ is asserted, the factor graph enforces the constraint in Eq. 2, which assumes the foothold is stationary. If this assumption is violated (e.g., due to slip), the optimization introduces errors in the estimated trajectory.

D. Velocity-Based Contact Refinement

Our approach refines contact sequences by exploiting a key observation: *true ground contacts should exhibit near-zero foot velocity in the world frame*. Given an initial trajectory estimate $\{\mathbf{x}_t\}$ from the factor graph, we can compute foot velocities using forward kinematics:

$$\mathbf{v}_t^{i,\text{FK}} = \mathbf{v}_t + \boldsymbol{\omega}_t \times \mathbf{r}_t^i + \dot{\mathbf{r}}_t^i \tag{4}$$

where $\mathbf{r}_t^i$ is the foot position in body frame from forward kinematics, $\boldsymbol{\omega}_t$ is angular velocity, and $\dot{\mathbf{r}}_t^i$ is the foot velocity in body frame.

E. Non-Convexity of Contact Selection

Ideally, contact selection would be optimized jointly with the robot trajectory inside the smoothing problem. In this case, the estimator would solve for both the continuous robot states and the discrete contact sequence. Let $\mathbf{x}_{0:T}$ denote the robot trajectory and let $c_t^i \in \{0, 1\}$ denote whether foot i is in contact at time t. A simplified contact-aware smoothing objective can be written as

$$\min_{\mathbf{x}_{0:T}, \mathbf{l}_{1:F}, c_{1:T}^{1:F}} \sum_{t=1}^{T-1} \|\mathbf{r}_{\text{imu}}(\mathbf{x}_t, \mathbf{x}_{t+1})\|^2_{\boldsymbol{\Sigma}_{\text{imu}}} + \sum_{t=1}^{T} \sum_{i=1}^{F} c_t^i \, \|\mathbf{r}_{\text{contact}}(\mathbf{x}_t, \mathbf{l}_i)\|^2_{\boldsymbol{\Sigma}_{\text{contact}}}, \tag{5}$$

subject to

$$c_t^i \in \{0, 1\}. \tag{6}$$

Here, $\mathbf{r}_{\text{imu}}$ is the IMU residual between consecutive states, and $\mathbf{r}_{\text{contact}}$ is the contact residual. For example, using the landmark-style contact measurement model,

$$\mathbf{r}_{\text{contact}}(\mathbf{x}_t, \mathbf{l}_i) = \mathbf{z}_t^i - \mathbf{R}_t^T(\mathbf{l}_i - \mathbf{p}_t). \tag{7}$$

This formulation captures the desired behavior: when $c_t^i = 1$, the estimator enforces the contact constraint; when $c_t^i = 0$, the contact residual is ignored. However, solving this problem directly is difficult because the contact variables and the trajectory variables are coupled.

There are several sources of non-convexity. First, the contact variables are binary, so the feasible set $c_t^i \in \{0, 1\}$ is non-convex. Second, the term

$$c_t^i \, \|\mathbf{r}_{\text{contact}}(\mathbf{x}_t, \mathbf{l}_i)\|^2 \tag{8}$$

creates a product between a discrete contact variable and a nonlinear residual depending on the robot state. This is a bilinear or mixed discrete-continuous coupling. Third, the residual itself is nonlinear because it depends on the robot orientation $\mathbf{R}_t$ and the transformation between the world frame and body frame. Therefore, even if the contact sequence were known, the smoothing problem would still be a nonlinear least-squares problem. If the contacts are also unknown, the estimator must search over both the continuous trajectory and the discrete contact schedule.

Because of these challenges, joint contact-and-state smoothing becomes a mixed-integer nonlinear optimization problem. This is a very difficult class of problem to convexify, and is generally intractable for real-time applications.

Our main idea is to run contact refinement and trajectory smoothing as an alternating outer-loop procedure. Starting from an initial, possibly noisy contact sequence, we first solve a standard factor graph smoothing problem to estimate the robot trajectory $\{\mathbf{x}_t\}$ and foothold positions $\{\mathbf{l}_i\}$. We then use this trajectory to compute foot velocities and solve a convex optimization problem that refines the contact sequence using velocity consistency, contact priors, and temporal smoothness.

The refined contact sequence is passed back into the smoother, and the process is repeated until the contacts change only minimally between iterations. This avoids directly solving the fully coupled contact-and-trajectory optimization problem, which is nonlinear and mixed discrete-continuous. Instead, the method alternates between two easier subproblems: nonlinear smoothing with fixed contacts, and convex contact refinement with a fixed trajectory.

III. Convex Reformulation / Convexification

A. Continuous Relaxation of Contact States

We relax the binary contact variables to the continuous interval:

$$c_t^i \in [0,1] \quad \forall t, i. \tag{9}$$

Here, $c_t^i = 1$ represents full contact, $c_t^i = 0$ represents no contact, and intermediate values represent uncertainty in the contact state. Rather than applying a single hard threshold at 0.5, we convert the relaxed contact values back to binary contacts using hysteresis thresholding.

Specifically, we use two thresholds, θ_{on} and θ_{off}, with

$$\theta_{\text{on}} = 0.6, \qquad \theta_{\text{off}} = 0.4. \tag{10}$$

The binary contact estimate is updated according to

$$\hat{c}_t^i = \begin{cases} 1, & c_t^i \geq \theta_{\text{on}}, \\ 0, & c_t^i \leq \theta_{\text{off}}, \\ \hat{c}_{t-1}^i, & \theta_{\text{off}} < c_t^i < \theta_{\text{on}}. \end{cases} \tag{11}$$

This hysteresis rule prevents rapid switching when the relaxed contact value is near the decision boundary.

B. Velocity-Based LP Formulation

Our key insight is to formulate the contact refinement objective such that contact is encouraged when the foot velocity is below a threshold. We define the velocity residual for foot i at time t as:

$$\rho_t^i = \|\mathbf{v}_t^{i,\text{FK}}\|_2 - r_0 \tag{12}$$

where $\|\mathbf{v}_t^{i,\text{FK}}\|_2$ is the norm of the foot velocity computed from forward kinematics (Eq. 4), and r_0 is a velocity threshold parameter (typically 0.2 m/s).

We incorporate this directly into the objective as:

$$\lambda_V \sum_{t=1}^{T} \rho_t^i \cdot c_t^i = \lambda_V \sum_{t=1}^{T} (\|\mathbf{v}_t^{i,\text{FK}}\|_2 - r_0) \cdot c_t^i \tag{13}$$

This formulation has a key property: when $\|\mathbf{v}_t^{i,\text{FK}}\|_2 < r_0$ (low velocity), the coefficient ρ_t^i is negative, so the term contributes a negative cost, effectively *rewarding* the optimizer for setting $c_t^i = 1$. Conversely, when $\|\mathbf{v}_t^{i,\text{FK}}\|_2 > r_0$ (high velocity), ρ_t^i is positive, penalizing contact. This naturally encourages $c_t^i = 1$ when the foot is stationary and $c_t^i = 0$ when moving.

Since the velocity norms $\|\mathbf{v}_t^{i,\text{FK}}\|_2$ are precomputed from the current smoothed trajectory and are not decision variables, this term remains linear in the decision variable c_t^i.

C. Prior and Smoothness Regularization

To maintain consistency with the initial contact sequence $\bar{c}_t^i$ (from the baseline detector), we introduce a prior term using epigraph variables $r_t \geq 0$:

$$r_t \geq c_t^i - \bar{c}_t^i \tag{14}$$
$$r_t \geq \bar{c}_t^i - c_t^i \tag{15}$$

This forces $r_t \geq |c_t^i - \bar{c}_t^i|$, penalizing deviations from the prior.

For temporal smoothness, we use total variation (TV) regularization with epigraph variables $d_t \geq 0$:

$$d_t \geq c_t^i - c_{t-1}^i \tag{16}$$
$$d_t \geq c_{t-1}^i - c_t^i \tag{17}$$

This enforces $d_t \geq |c_t^i - c_{t-1}^i|$, discouraging rapid contact switching.

D. Complete Convex Problem Formulation

For each foot i independently, we solve the following LP:

$$\min_{c_{1:T}^i, r, d} \quad \lambda_V \sum_{t=1}^{T} (\|\mathbf{v}_t^{i,\text{FK}}\|_2 - r_0) c_t^i + \beta \sum_{t=1}^{T} r_t + \lambda_{\text{TV}} \sum_{t=2}^{T} d_t \tag{18}$$

$$\text{subject to} \quad r_t \geq c_t^i - \bar{c}_t^i \quad \forall t \tag{19}$$
$$r_t \geq \bar{c}_t^i - c_t^i \quad \forall t \tag{20}$$
$$d_t \geq c_t^i - c_{t-1}^i \quad \forall t \geq 2 \tag{21}$$
$$d_t \geq c_{t-1}^i - c_t^i \quad \forall t \geq 2 \tag{22}$$
$$0 \leq c_t^i \leq 1 \quad \forall t \tag{23}$$
$$r_t, d_t \geq 0 \quad \forall t \tag{24}$$

where:

- $\lambda_V > 0$: Weight for velocity-based contact term (can reward or penalize contact)
- $r_0 > 0$: Velocity threshold parameter (0.3m/s in our experiments)
- $\beta > 0$: Weight for prior deviation
- $\lambda_{\text{TV}} > 0$: Weight for total variation smoothness

IV. Solving the Convex Problem

A. Solver Selection and Implementation

We implement the system using:

- **GTSAM** (Georgia Tech Smoothing and Mapping): Factor graph optimization framework with fixed-lag smoothing
- **CVXPY**: Convex optimization modeling language (Python)
- **SciPy/HiGHS** [3]: High-performance linear programming solver via SciPy's optimization interface

For each foot, the LP has:

- Decision variables: $3T$ (contact states c_t, epigraphs r_t, d_t)
- Linear constraints: $4T$ (epigraph bounds for prior and TV regularization)

Typical solve time: 0.05 seconds for $T = 500$ timesteps.

B. Outer-Loop Refinement Algorithm

Our complete estimation pipeline consists of two nested loops:

Algorithm 1: Contact-Refined Factor Graph Smoothing

Input: IMU data $\{\mathbf{u}_t\}$, initial contact sequence $\{\bar{c}_t^i\}$, parameters $(r_0, \lambda_V, \beta, \lambda_{\text{TV}})$
Output: Refined trajectory $\{\mathbf{x}_t^*\}$, refined contacts $\{\hat{c}_t^i\}$

// Inner loop: Factor graph optimization
Initialize factor graph with initial contacts $\{\bar{c}_t^i\}$
Add IMU preintegration factors
Add contact measurement factors for timesteps where $\bar{c}_t^i = 1$
Solve MAP inference: $(\mathbf{x}_{1:T}, \mathbf{l}_{1:F}) \leftarrow \arg\max p(\cdot|\mathbf{z}, \mathbf{u})$

// Compute foot velocities from estimated trajectory
for $t = 1$ to T **do**
  for each foot i **do**
    Compute $\mathbf{v}_t^{i,\text{FK}}$ using forward kinematics (Eq. 4)
    Compute velocity norm: $\|\mathbf{v}_t^{i,\text{FK}}\|_2$
  end for
end for

// Outer loop: Convex contact refinement
for each foot $i = 1$ to F **do**
  Formulate LP with velocity norms $\{\|\mathbf{v}_t^{i,\text{FK}}\|_2\}$
  Solve LP using CVXPY + SciPy/HiGHS solver
  Apply hysteresis thresholding: $\hat{c}_t^i = 1$ if $c_t^i \geq 0.6$, $\hat{c}_t^i = 0$ if $c_t^i \leq 0.4$, else $\hat{c}_t^i = \hat{c}_{t-1}^i$
end for

// Check convergence: Repeat until max iterations or change in contact sequence $< 5\%$
if contacts changed $< 5\%$ or max iterations reached **then**
  break
end if
// Re-optimize factor graph with refined contacts (optional)
Rebuild factor graph with refined contacts $\{\hat{c}_t^i\}$
Solve MAP inference again to get updated trajectory $\{\mathbf{x}_t^*\}$

return $(\{\mathbf{x}_t^*\}, \{\hat{c}_t^i\})$

V. Comparison and Critical Evaluation

A. Baseline Methods

We compare the proposed convex contact refinement method against two baselines:

1) **Threshold-only**: Simple velocity thresholding without optimization (if $\|\mathbf{v}_t^i\| < \epsilon$, then contact)
2) **Factor graph without refinement**: GTSAM fixed-lag smoother with original contact detections

B. Evaluation Metrics

We evaluate methods on:

- **Trajectory accuracy**: RMS position, rotation, and velocity vs. ground truth (from simulation)
- **Computation time**: Total optimization time

C. Experimental Results

Table I summarizes performance across three bipedal robot datasets comparing our convex optimization approach against baseline factor graph smoothing and simple threshold-based contact detection. For velocity thresholding, we set the foot velocity threshold as 0.3m/s Beyond this threshold, the contact is turned off. The weighting parameters for the convex optimization were set as $\lambda_V = 3$, $\beta = 8$, and $\lambda_{\text{TV}} = 1$, and our residual threshold was set as $r_0 = 0.3$ m/s.

The average time taken for the LP to be solved was 0.05 seconds per 500 samples on average.

D. Critical Analysis

Advantages of convex refinement:

- **Global optimality**: LP guarantees global optimum (within continuous relaxation)
- **Principled trade-offs**: Intuitive weighting of velocity residual penalty vs. prior consistency vs. smoothness. All of them share the same degree of magnitude.

Limitations:

- **Relaxation gap**: Continuous relaxation may not perfectly recover binary contacts (hysteresis thresholding introduces approximation)
- **Velocity dependence**: Method assumes factor graph produces reasonable foot velocity residuals
- **Computational overhead**: Factor graphs smoothing take approximately 1 second for 500 samples of data, resulting in a 5% increase in total pipeline time compared to baseline factor graph
- **Parameter tuning**: Weights $(\lambda_V, \beta, \lambda_{\text{TV}})$ and threshold r_0 require problem-specific adjustment

Trade-offs: The method achieves substantial improvements across all datasets. Compared to baseline factor graph smoothing, our convex refinement achieves 39.9% average improvement in position accuracy and 22.5% in velocity consistency. Against simple threshold-based detection, improvements are

TABLE I: Performance Comparison: Convex Refinement vs. Baseline Methods

Dataset	Comparison	Ref. Pos. RMSE (m)	Convex Pos. RMSE (m)	Pos. Imp. (%)	Ref. Vel. RMSE (m/s)	Convex Vel. RMSE (m/s)	Vel. Imp. (%)
Dataset 1	vs. Baseline	0.3397	0.3108	8.5	0.2728	0.2278	16.5
	vs. Thresholded	0.8532	0.3108	63.6	0.3457	0.2278	34.1
Dataset 2	vs. Baseline	0.8910	0.0977	89.0	0.7539	0.4732	37.2
	vs. Thresholded	3.7011	0.0977	97.4	2.0504	0.4732	76.9
Dataset 3	vs. Baseline	0.5845	0.4551	22.1	0.5997	0.5178	13.7
	vs. Thresholded	5.6583	0.4551	92.0	0.9293	0.5178	44.3
Aggregate Performance							
Avg. vs. Baseline		–	–	39.9	–	–	22.5
Avg. vs. Thresholded		–	–	84.3	–	–	51.8

Note: All datasets are from Unitree G1 robot.

even more dramatic: 84.3% for position and 51.8% for velocity. The computational overhead remains modest (5% increase in total pipeline time).

VI. Discussion of Cost Formulation and Alternatives

The velocity-based contact cost plays a central role in the proposed convex contact smoothing framework. The objective of this term is to encourage contact when the estimated foot velocity is sufficiently small, while discouraging contact when the velocity becomes large. Several formulations were explored before selecting the final residual structure.

A. Linear Signed Residual Formulation

The selected formulation uses a signed linear residual of the form

$$J_v = \sum_{t,i} w_v \left(r_t^i - r_0\right) c_t^i,$$

where:

- r_t^i is the velocity residual (or foot velocity magnitude),
- r_0 is a reference velocity threshold,
- $c_t^i \in [0, 1]$ is the relaxed contact variable,
- w_v is the weighting coefficient.

This formulation provides a simple and interpretable mechanism for encouraging or discouraging contact:

- If $r_t^i < r_0$, then $(r_t^i - r_0) < 0$, and the optimizer lowers the objective by increasing c_t^i, thereby encouraging contact.
- If $r_t^i > r_0$, then $(r_t^i - r_0) > 0$, and the optimizer lowers the objective by decreasing c_t^i, thereby discouraging contact.

A major advantage of this formulation is that it remains linear in the decision variable c_t^i, preserving computational efficiency. Additionally, the linear residual naturally encourages c_t^i toward binary values rather than fractional values, which is desirable since the final contact decisions are ultimately thresholded to binary contact states. Furthermore, the residual maintains proportionality with the velocity magnitude, meaning that larger violations produce proportionally larger penalties.

B. Sigmoid Probability Formulation

An alternative formulation considered mapping the velocity residual into a probability using a sigmoid function:

$$p_t^i = \sigma\left(k(r_t^i - r_0)\right),$$

where

$$\sigma(x) = \frac{1}{1 + e^{-x}},$$

and k controls the sharpness of the transition.

The resulting probability could then be incorporated into the optimization through a consistency cost such as

$$J_v = \sum_{t,i} \left(c_t^i - p_t^i\right)^2.$$

The primary motivation for this approach is that it provides a probabilistic interpretation of the velocity evidence. Velocities below the threshold map to high contact confidence, while larger velocities map to low contact confidence. The parameter k controls the sharpness of the transition.

However, several practical limitations were observed:

- The formulation introduces additional tuning sensitivity through both k and r_0.
- Sigmoid functions saturate for large residual magnitudes, reducing sensitivity to large violations.
- The formulation naturally encourages fractional contact values near the transition region, despite the final contact decisions ultimately being thresholded to binary values. This is not exactly desired behavior for us in this scenario

C. Quadratic Residual Formulation

Another alternative considered was a quadratic residual formulation:

$$J_v = \sum_{t,i} w_v \left(r_t^i - r_0\right)^2 c_t^i.$$

This formulation penalizes large deviations from the velocity threshold more aggressively than the linear residual.

One advantage of the quadratic formulation is that it produces smoother gradients and stronger suppression of large velocity violations. It may therefore be beneficial in scenarios where severe contact violations must be strongly discouraged.

However, the quadratic residual also introduces several drawbacks:

- Large residuals can dominate the optimization objective, especially in the presence of noisy velocity estimates.
- The formulation loses the directional linear behavior of the signed residual.

Compared to the quadratic formulation, the selected signed linear residual provides a more balanced tradeoff between interpretability, convexity, robustness, and optimization stability.

VII. Conclusion

This paper presented a convex optimization framework for refining contact sequences in factor graph-based state estimation for bipedal robots. By treating footholds as landmarks and formulating contact refinement as a velocity-constrained LP, we achieved significant improvements in estimation accuracy while maintaining computational efficiency.

The key contributions are: (1) A convex LP that globally optimizes contact sequences via continuous relaxation, (2) Integration of the refinement loop with GTSAM's fixed-lag smoother framework, and (3) Experimental validation demonstrating 39.9% average improvement in position accuracy and 22.5% in velocity consistency compared to baseline factor graph smoothing, with up to 89.0% improvement on challenging datasets. Against simple threshold-based methods, our approach achieves 84.3% and 51.8% average improvements in position and velocity metrics, respectively.

The outer-loop refinement approach is modular and can be integrated with any factor graph backend (not just GTSAM). The convex formulation avoids local minima issues of greedy methods while providing interpretable parameters for trading off velocity consistency, prior fidelity, and temporal smoothness.

VIII. Future Work

One possible extension of this work is to optimize contact covariances rather than only optimizing the contact states themselves. In the current framework, the contact covariance used during factor graph smoothing remains fixed throughout optimization. However, contact confidence could instead be used to adaptively scale the covariance of the contact factors, allowing uncertain contacts to contribute less strongly to the smoothing process.

For such a formulation, probabilistic residual models such as sigmoid-based contact confidence mappings may become more useful due to their continuous nature. Rather than producing only binary contact decisions, these formulations could provide smooth confidence estimates that directly modulate contact uncertainty during smoothing.

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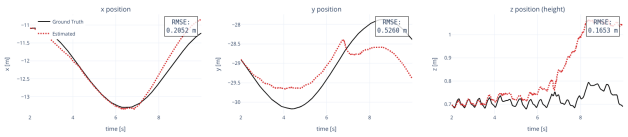
Appendix

This appendix presents detailed trajectory estimation results for all three datasets, comparing the baseline factor graph smoother, threshold-based contact detection, and the proposed convex optimization approach.

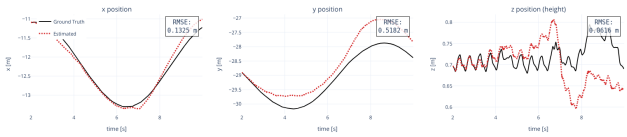
A. Dataset 1

B. Dataset 2

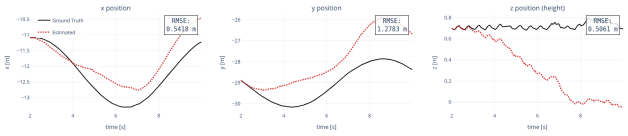
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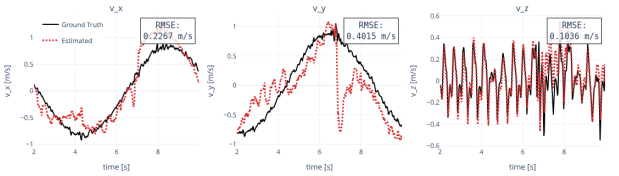
(a) Baseline - Position (RMSE: x=0.21m, y=0.53m, z=0.17m)



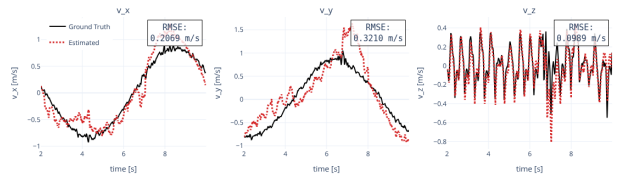
(b) Convex - Position (RMSE: x=0.06m, y=0.05m, z=0.15m)



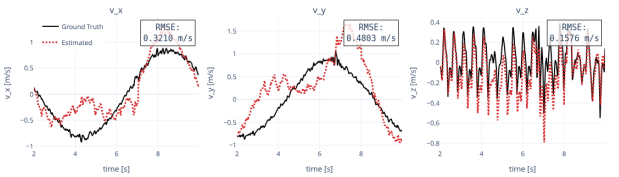
(c) Threshold - Position (RMSE: x=5.79m, y=3.95m, z=6.85m)



(d) Baseline - Velocity (RMSE: x=0.23m/s, y=0.40m/s, z=0.10m/s)

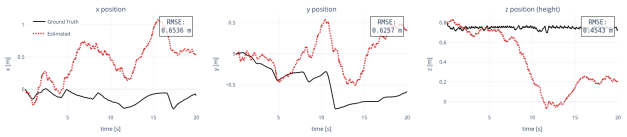


(e) Convex - Velocity (RMSE: x=0.21m/s, y=0.32m/s, z=0.10m/s)

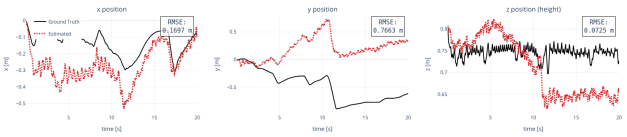


(f) Threshold - Velocity (RMSE: x=0.53m/s, y=0.59m/s, z=0.42m/s)

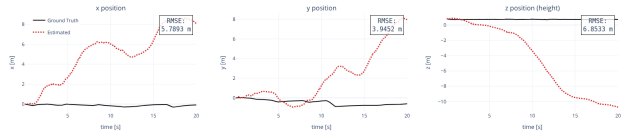
Fig. 1: Dataset 1 trajectory estimation results comparing position and velocity estimates across three methods: baseline factor graph smoothing, proposed convex optimization, and threshold-based contact detection.



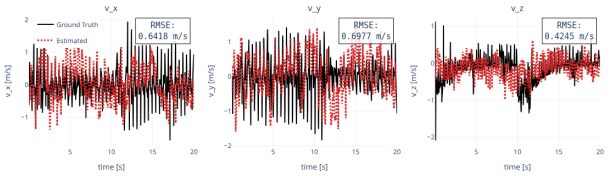
(a) Baseline - Position (RMSE: x=0.65m, y=0.63m, z=0.45m)



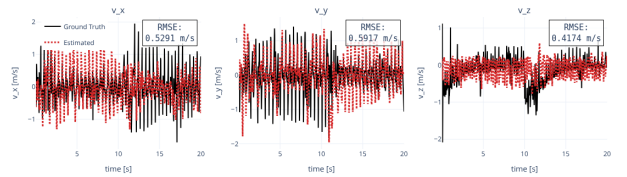
(b) Convex - Position (RMSE: x=0.17m, y=0.77m, z=0.07m)



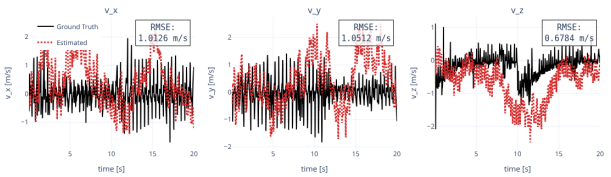
(c) Threshold - Position (RMSE: x=0.56m, y=6.23m, z=1.42m)



(d) Baseline - Velocity (RMSE: x=0.64m/s, y=0.70m/s, z=0.42m/s)

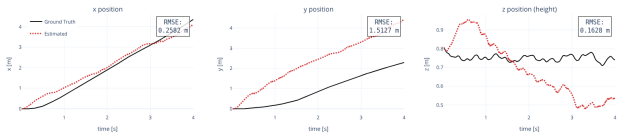


(e) Convex - Velocity (RMSE: x=0.32m/s, y=0.48m/s, z=0.16m/s)

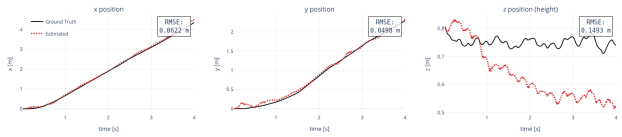


(f) Threshold - Velocity (RMSE: x=0.95m/s, y=3.28m/s, z=0.98m/s)

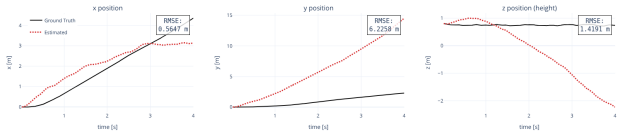
Fig. 2: Dataset 2 trajectory estimation results comparing position and velocity estimates across three methods: baseline factor graph smoothing, proposed convex optimization, and threshold-based contact detection.



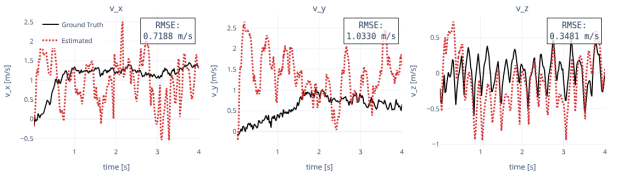
(a) Baseline - Position (RMSE: x=0.26m, y=1.51m, z=0.16m)



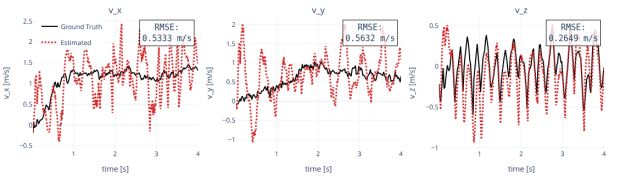
(b) Convex - Position (RMSE: x=0.54m, y=1.28m, z=0.51m)



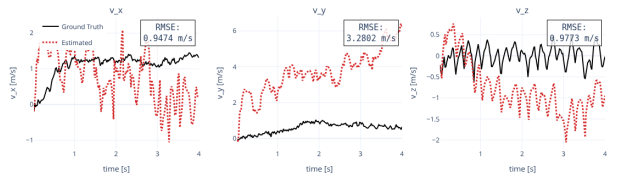
(c) Threshold - Position (RMSE: x=0.56m, y=6.23m, z=1.42m)



(d) Baseline - Velocity (RMSE: x=0.72m/s, y=1.03m/s, z=0.35m/s)



(e) Convex - Velocity (RMSE: x=0.53m/s, y=0.56m/s, z=0.26m/s)



(f) Threshold - Velocity (RMSE: x=1.01m/s, y=1.05m/s, z=0.68m/s)

Fig. 3: Dataset 3 trajectory estimation results comparing position and velocity estimates across three methods: baseline factor graph smoothing, proposed convex optimization, and threshold-based contact detection.